

Conformal Calibration of Open-Vocabulary Semantic Maps under Label and Pose Uncertainty

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Abstract—A robot that avoids “the plant” on an open-vocabulary semantic map is exposed to two errors at once. The detector misses, truncates or mislabels objects, and the pose estimate drifts, so objects are drawn in the wrong place. Existing distribution-free guarantees for planning on perception maps calibrate the first error and assume a known pose. We calibrate one per-scene quantity, the *miss distance*: the directed Hausdorff distance from the true footprints of a class to the class region on the map built with estimated poses, measured in the planner’s frame. Its split-conformal quantile is a keep-out margin. Any path that stays this margin plus the safety distance away from the class region is safe with the target probability, whichever planner produced it. On 21 OSMa-Bench scenes the joint margin keeps 0.93 coverage at a 0.9 target from zero to 45 cm of drift, while margins calibrated on labels or on pose alone lose coverage. In the setting of the closest prior work, known pose and a closed vocabulary, label-set calibration and ours achieve the same mission success. With drift or an open vocabulary ours keeps its guarantee and succeeds more often. With real visual odometry and no loop closure the method abstains instead of issuing a false certificate.

I. INTRODUCTION

Open-vocabulary semantic maps [1] let a robot follow instructions such as “keep away from the bike”. The map has two unreliable inputs. A detector or segmenter supplies the labels, and it misses objects, covers them only in part, or confuses classes. The robot’s pose estimate places the observations, and it drifts. Planners usually take the map at face value and keep a fixed clearance from the regions of avoid classes.

How unsafe this is usually goes unmeasured. OSMa-Bench [2] shows that open-vocabulary maps degrade sharply under lighting changes (the f-mIoU of ConceptGraphs drops from 28.00 to 14.29), but it scores the map as an image, without a robot moving through it. A recent survey of 3D scene graphs [3] lists the link between map errors and robot behaviour as an open problem.

Conformal prediction (CP) [4] gives a planner guarantees without assumptions on the error distribution. Sundarsingh et al. [5] calibrate per-cell label sets of a semantic grid; Perceive With Confidence [6] calibrates how much to inflate an occupancy map. Both assume that the robot’s pose is known, and separately calibrated modules need not form a calibrated system [7].

We ask what guarantee a planner gets when label error and pose error act together, and how to calibrate it. Our contributions are:

- a nonconformity score, the miss distance, measured in the planner frame on a map built with estimated poses; its per-class conformal quantile is a margin that covers both errors for any planner;
- an evaluation on 21 OSMa-Bench scenes with six levels of synthetic drift and real visual odometry, which shows that margins calibrated on one error lose coverage under the other and that the sum of two separate margins either over-pays or under-covers, depending on how the errors are distributed over scenes;
- a comparison with label-set calibration [5] in its own setting and in ours, which explains why label-space calibration degenerates on indoor maps.

II. RELATED WORK

Open-vocabulary mapping. ConceptGraphs [1] fuses class-agnostic masks into an object graph with CLIP features; OSMa-Bench [2] and OSMa-Bench++ [8] evaluate such mappers under lighting changes. Naive Bayesian fusion makes semantic maps over-confident [9]; we fuse by averaging. None of these works evaluates the map through a motion task.

Conformal guarantees for planning. CP entered planning through trajectory prediction of other agents [10]. Shin et al. [11] conformalize the whole distance field to moving obstacles and obtain a constraint that certifies any MPC trajectory; semantics and the robot’s own pose error are outside their scope. Learnable CP [12] learns a context-aware score for RRT* clearance, and constraint tightening [13] shows that tightened constraints are safe for any planner. Closest to us, Sundarsingh et al. [5] calibrate label sets of grid cells with one score per environment (the worst cell) and plan with the most conservative label of each cell. They report 93.4% correct maps and 98.4% mission success at a 95% target, with a known pose, five classes and a 1 m grid on 70×25 m outdoor sites. Perceive With Confidence [6] calibrates the smallest inflation of an occupancy map that covers all obstacles; our score is a semantic counterpart of theirs that also absorbs pose error.

Composition and pose error. Two modules calibrated at 95% give between 86 and 99.98% after composition depending on error correlation, and joint calibration restores the target [7]. Calafiore [14] combines blockwise certificates through a union bound with an explicit risk budget; in his example a joint calibration of the supremum residual gives a slightly tighter margin. Pose error in semantic memory is real and poorly calibrated: on a real robot the nominal 95% pose

ellipsoid covers a third of the actual errors [15]. We found no prior work that calibrates open-vocabulary label error and pose error jointly with a planner in the loop; the claim rests on a citation and keyword search of October 2026.

III. PROBLEM FORMULATION

A scene Ω is drawn from an unknown distribution $\mathcal{D}$. It contains objects of avoid classes $\mathcal{K}$; object e has class k_e and floor footprint $F_e \subset \mathbb{R}^2$, and $F_k = \bigcup_{e: k_e=k} F_e$. The robot moves along true poses $T_1, \dots, T_q \in SE(2)$, and its estimator returns $\hat{T}_t = D_t T_t$ with accumulated error $D_t \in SE(2)$, $D_1 = I$. A grid map $\hat{M}$ is fused from all observations placed with the *estimated* poses; cell j stores the score-weighted fraction $p_j(k)$ of its observations that support class k . The region of class k at threshold λ is

$$A_\lambda(k) = \{j : j \text{ occupied, } p_j(k) \geq \lambda\}. \quad (1)$$

The planner acts at time q in the world as the robot believes it to be. An object seen at time t is drawn with error D_t , while the robot sits with error D_q , so in the planner frame the true footprint is $\hat{F}_k = D_q F_k$. A constant pose error moves the map and the robot together and is harmless; the difference between the error at observation time and at query time is what hurts.

A path $\gamma: [0, 1] \rightarrow \mathbb{R}^2$ is safe for class k if $\text{dist}(\gamma(\tau), \hat{F}_k) \geq d$ for all τ . Given n calibration scenes with known ground truth, we seek per-class margins r_k that are as small as possible and satisfy, for a new scene,

$$\mathbb{P} \left(\begin{array}{l} \text{every path with } \text{dist}(\gamma, A_\lambda(k)) \geq d + r_k \\ \text{is safe for class } k \end{array} \right) \geq 1 - \alpha. \quad (2)$$

The probability is over the calibration scenes, the new scene and the error realizations.

Assumption 1 (exchangeability). The calibration scenes $\Omega_1, \dots, \Omega_n$ and the new scene Ω_{n+1} are exchangeable.

No model of the detector or of the pose estimator is needed. We test four hypotheses: a margin calibrated at the true pose loses coverage once drift is comparable to the label error (H1); a margin calibrated with ground-truth labels undercovers while drift is smaller than the label error (H2); the joint margin of Sec. IV keeps coverage at every drift level and is smaller than the sum of the two (H3); map mIoU is a poor predictor of mission safety (H4).

IV. METHOD

Definition 1 (miss distance). For scene Ω and class k ,

$$s_k(\Omega) = \min\{R_{\max}, h(\hat{F}_k, A_\lambda(k))\}, \quad (3)$$

where $h(X, Y) = \sup_{x \in X} \inf_{y \in Y} \|x - y\|$ is the directed Hausdorff distance from the true footprints to the class region; $s_k = 0$ if the scene has no object of class k and $s_k = R_{\max}$ if the region is empty.

A missed or mislabeled object gives a large score, a truncated object gives the length of the missing part, and a pose shift gives the size of the shift. Both errors enter one number in the way they affect the planner. We compute one score

per scene: cells, frames and objects of one scene are strongly dependent and are not exchangeable units.

Calibration. Let $s^{(1)} \leq \dots \leq s^{(n)}$ be the sorted calibration scores of class k and $m = \lceil (n+1)(1-\alpha) \rceil$. The margin is $\hat{r}_k = s^{(m)}$. If $m > n$ (that is, $n < 1/\alpha - 1$) or $\hat{r}_k \geq R_{\max}$, the class *abstains*: its certificate is vacuous. Calibrating per class keeps one unreliable class from inflating the others.

Proposition 1. Under Assumption 1, $\mathbb{P}(s_k(\Omega_{n+1}) \leq \hat{r}_k) \geq 1 - \alpha$, and $r_k = \hat{r}_k$ satisfies (2).

Proof. By exchangeability the rank of $s_k(\Omega_{n+1})$ among the $n+1$ scores is uniform (ties broken in favour of coverage), so $\mathbb{P}(s_k(\Omega_{n+1}) \leq s^{(m)}) \geq m/(n+1) \geq 1 - \alpha$ [4]. If $s_k \leq \hat{r}_k$, every $y \in \hat{F}_k$ has some $a \in A_\lambda(k)$ with $\|y - a\| \leq \hat{r}_k$. For a path point x with $\text{dist}(x, A_\lambda(k)) \geq d + \hat{r}_k$, the triangle inequality gives $\|x - y\| \geq \|x - a\| - \|a - y\| \geq d$. $\square$

The certificate does not depend on the planner; the planner only decides whether a path exists in the narrowed free space. With $n = 13$ and $\alpha = 0.1$, $m = 13$ and the margin is the largest calibration score.

Compared calibrations. All arms use the same region $A_{\lambda_0}(k)$, with $\lambda_0 = 0.05$ chosen on a development scene, and differ in the scores the margin is calibrated on: *uncalibrated* (margin 0); *labels only*, score (3) on the detector map at true poses, giving r_k^{lab} ; *pose only*, the same score on a ground-truth-label map built with drifted poses, giving r_k^{pose} ; *separate*, $r_k^{\text{lab}} + r_k^{\text{pose}}$ with both at level α ; and *joint*, the score on the detector map built with drifted poses. We also re-implement *label sets* [5]: the score is the worst $1 - p_j(k)$ over true footprint cells, the threshold is $\lambda^* = 1 - \hat{q}$ with $\hat{q}$ its conformal quantile, and every cell whose label set contains k is avoided.

Proposition 2. If $s_k \leq s_k^{\text{lab}} + s_k^{\text{pose}}$ for the scores of the same scene, the separate margin has coverage at least $1 - 2\alpha$, not $1 - \alpha$.

Proof. Each margin fails to cover its own score with probability at most α . By the union bound both cover with probability at least $1 - 2\alpha$, and then $s_k \leq r_k^{\text{lab}} + r_k^{\text{pose}}$. $\square$

A risk budget of $\alpha/2$ per module [14] restores $1 - \alpha$ but needs $n \geq 2/\alpha - 1$, that is 19 calibration scenes at $\alpha = 0.1$. Where the sum lands between $1 - 2\alpha$ and 1 depends on the dependence of the two errors, which is unknown in practice. The joint margin does not depend on it.

Missions. A robot of radius 0.2 m must reach a goal. The keep-out zone of class k is $A_{\lambda_0}(k)$ dilated by $d + r_k$, with $d = 0.5$ m. Dijkstra's algorithm on the 8-connected grid finds the shortest path; it is complete, so "no path" means that none exists. A path is *unsafe* if it comes closer than d to a true avoid object in the planner frame; a mission *succeeds* if a safe path is found. The same planner on the true map serves as an oracle.

V. EXPERIMENTAL SETUP

Data and perception. We use 22 ReplicaCAD scenes of OSMA-Bench [2], [16] with RGB-D, ground-truth semantics

TABLE I
COVERAGE AND MEDIAN MARGIN (M) FOR BIKE AND PLANT, $1 - \alpha = 0.9$.
MEDIAN ATE: L2 2 cm, L4 45 cm. [†] THE ARM ABSTAINS IN AT LEAST HALF OF THE SPLITS; THE MARGIN OF THE OTHERS IS SHOWN.

calibration	bike				indoor plant			
	L0	L2	L4	$\hat{r}$ L4	L0	L2	L4	$\hat{r}$ L4
uncalibrated	0.00	0.00	0.01	0	0.00	0.00	0.00	0
label sets [5]	1.00	0.73	0.17	—	1.00	0.41	0.03	—
labels only	0.93	0.94	0.76	0.75	0.93	0.96	0.86	0.85 [†]
pose only	0.00	0.03	0.82	0.90	0.00	0.00	0.86	1.39
separate	0.93	0.98	0.97	1.61	0.93	0.98	0.95	2.11 [†]
joint (ours)	0.93	0.94	0.93	1.18	0.93	0.93	0.93	1.53[†]

and poses at 640×480 , every 8th frame. All design choices were made on apt_0; the other 21 scenes are split 200 times at random into 13 calibration and 8 test scenes. YOLO-World v2-s [17] runs with the 98 object classes of the dataset as prompts; inside each box we keep pixels within 0.25 m of the median box depth. Maps are 5 cm grids, and points 0.1–1.8 m above the floor are obstacles. The avoid classes are indoor plant, bike and TV stand.

Pose error. Synthetic drift replays every relative motion of the true trajectory with planar noise proportional to the distance travelled and the angle turned. Six levels are set once on the development scene by target absolute trajectory error (ATE). On the evaluation scenes, whose paths are shorter, the median ATE is 0.4 cm (L1), 2 cm (L2), 7 cm (L3), 45 cm (L4, 3.6% of the path) and 2.7 m (L5); we draw 10 realizations per level. As real pose error we run frame-to-frame RGB-D odometry of Open3D [18], [19] on every 2nd frame with a planar ground-robot constraint and no loop closure, without tuning it on ground truth.

Protocol and tasks. $\alpha = 0.1$ unless stated, $R_{\max} = 3$ m; abstention counts as vacuous coverage and is reported. Start and goal are at least 1.5 m from avoid objects and 3 m apart. *Hard* tasks are selected so that the shortest obstacle-free path passes within 0.5 m of a plant or bike (233 tasks in 15 scenes); *random* tasks are not selected (20 per scene). Margins for a scene are calibrated on all other scenes. A class that abstains falls back to treating every occupied cell as that class.

VI. RESULTS

A. Coverage under pose drift

Table I tests H1–H3. The uncalibrated map almost never contains the whole true footprint, because the detector truncates object edges. Label sets reach 1.00 at the true pose only through $\lambda^* = 0$, which declares every occupied cell possibly hazardous; with no metric slack, 2 cm of drift drops their coverage to 0.73. The labels-only margin holds while drift stays below the label error (up to 7 cm) and falls to 0.76 at 45 cm (H1). The pose-only margin cannot see detector errors and stays at or below 0.22 up to 7 cm (H2). The joint margin holds 0.93–0.94 at every level and at 45 cm is 27–28 % smaller than the sum of separate margins (H3).

The plant abstains in about 60 % of the splits because the detector misses it completely in one scene, and with 13

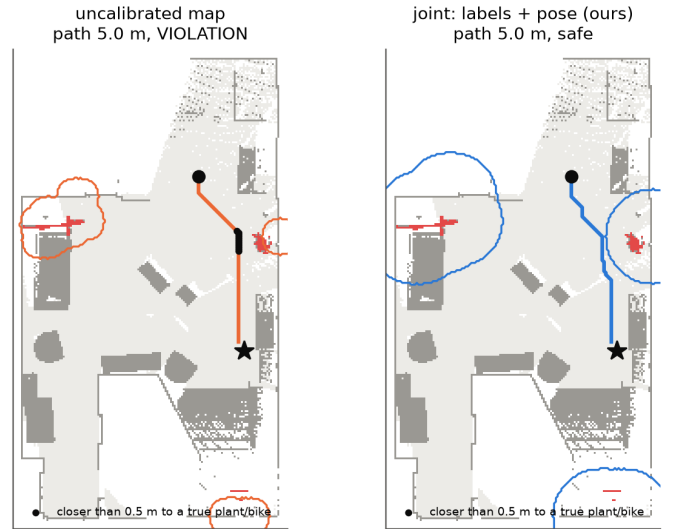


Fig. 1. A hard task at L2. Red: true plant and bike; line: keep-out boundary. The uncalibrated zone misses part of the plant and the path passes within 0.5 m of it (black dots); the joint zone covers it.

TABLE II
RANDOM TASKS: COVERAGE (MINIMUM OVER PLANT AND BIKE), MISSION SUCCESS AND UNSAFE MISSIONS.

setting	drift	label sets [5]			joint (ours)		
		cov.	succ.	unsafe	cov.	succ.	unsafe
closed vocab., $1 - \alpha = 0.95$	none	1.00	0.43	0.000	0.94	0.42	0.000
	2 cm	0.42	0.40	0.000	0.94	0.61	0.000
	7 cm	0.10	0.32	0.002	0.94	0.50	0.000
open vocab., $1 - \alpha = 0.9$	none	1.00	0.43	0.000	0.93	0.78	0.000
	2 cm	0.41	0.40	0.000	0.93	0.74	0.000
	7 cm	0.10	0.32	0.002	0.94	0.68	0.000

calibration scenes the margin is the maximum. The TV stand abstains in 71–95 % of the splits: in 2 of 21 scenes the detector confidently labels it as a bench or a rug. For this detector the TV stand cannot be certified at 90 %, and the calibration says so.

B. Missions

On hard tasks the uncalibrated planner almost always finds a path, but 27–37 % of its paths are unsafe at L0–L3 (Fig. 1). Every calibrated arm has no unsafe path at any level. The price is that a path exists in only 30–32 % of these deliberately hard tasks, against 98–100 % on the true map. On random tasks the joint margin gives 78 % successful missions at the true pose and 68 % at L3, with no unsafe one; the uncalibrated planner succeeds in 95 and 82 %, but 4–6 % of its missions are unsafe. Map quality does not predict safety: across scenes, the Spearman correlation between mIoU and the violation rate of the uncalibrated planner is -0.16 ($p = 0.57$, 15 scenes), which supports H4.

C. Comparison with label sets in their setting

The mission success of [5] comes from a different setting, so we compare the methods on our scenes under their assump-

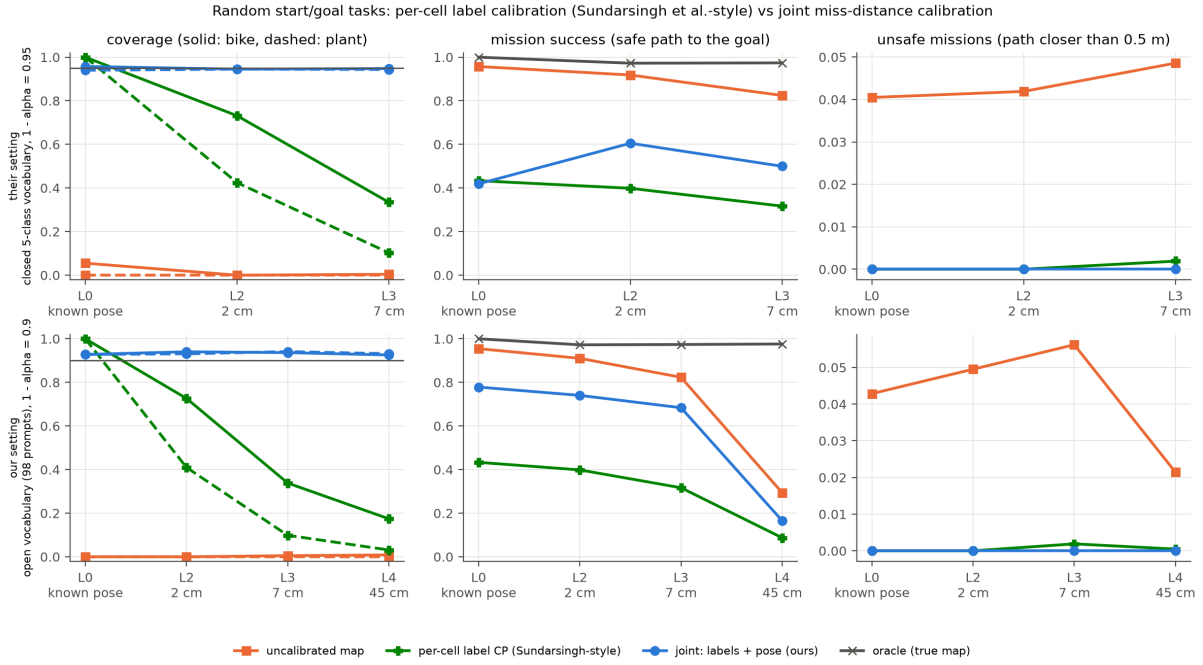


Fig. 2. Label sets [5] and the joint margin on random tasks. Top: closed vocabulary, $1 - \alpha = 0.95$; bottom: open vocabulary, $1 - \alpha = 0.9$.

tions: known pose, a closed vocabulary of five classes (five rows of the detector’s prompt embeddings, which is equivalent to re-prompting it), $1 - \alpha = 0.95$ and random tasks. Their exploration mode is not reproduced. In their setting the two calibrations are equal (0.43 vs. 0.42 mission success, no unsafe missions; Table II, Fig. 2). With 2 and 7 cm of drift the coverage of label sets falls to 0.42 and 0.10, while the joint margin stays at 0.94 and succeeds more often (0.61 and 0.50 vs. 0.40 and 0.32). With the open vocabulary the joint margin succeeds 1.8 times as often already at the true pose.

Why do label sets degenerate here? Their threshold is $\lambda^* = 0$ with both vocabularies. We suspected the fine grid and repeated both calibrations at the true pose on grids of 5, 25, 50 and 100 cm. The threshold is zero at every resolution, and the keep-out zone of label sets takes 64% of the free space at 5 cm and all of it at 1 m. The cause is complete misses: if in one calibration scene no observation of the class reaches part of the object, the worst cell scores 1 and the threshold drops to zero. A label set can only say “this cell may be a plant”. A distance says “a plant is nearby”, which stays informative when the detector loses parts of objects.

D. The price of the guarantee

Raising the risk level barely helps. From $\alpha = 0.1$ to 0.3 the joint margin shrinks by 0.2–0.25 m and the share of hard tasks with a path goes from 30 to 31 % at L3, while coverage tracks $1 - \alpha$ (0.94, 0.87, 0.72 at $\alpha = 0.1, 0.2, 0.3$). Better masks do not shrink the margin either. MobileSAM [20] prompted with the detector boxes cuts the median bike miss from 0.43 to 0.05 m, but the worst scene keeps a 0.75 m miss and, with 13 calibration scenes, sets the margin. With SAM masks the

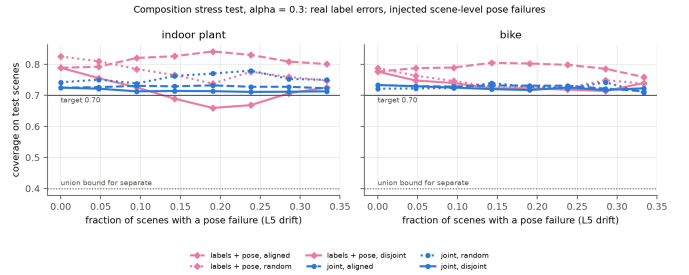


Fig. 3. Coverage of separate (pink) and joint (blue) margins vs. the fraction of scenes with a localization failure placed where labels are worst (dashed), best (solid) or at random (dotted); $1 - \alpha = 0.7$.

uncalibrated planner is safe at the true pose but not under drift (9.5% unsafe paths at L3). The margin is set by the tail of perception errors.

E. When does the sum of margins under-cover?

On synthetic drift, which is independent of detector errors, the sum of margins only over-pays (Table I). To probe Proposition 2 we keep the real label errors and replace drift by a scene-level mixture: in k of 21 scenes a localization failure occurs (realizations from L5), in the others drift is negligible (L1). Failures are placed in the scenes with the worst labels, the best labels, or at random. We use $\alpha = 0.3$, because at $\alpha = 0.1$ with 13 scenes any failure enters the maximum. The sum covers 0.76–0.84 when failures coincide with bad labels but only 0.66 at a 0.70 target when they hit scenes with good labels (plant, 19–24 % failing scenes). The joint margin stays at 0.71–0.78 however the failures are placed (Fig. 3).

Frame-to-frame odometry without loop closure errs in jumps, mostly during in-place turns near walls: its ATE ranges from 3.5 cm to 3.9 m across scenes, with a median of 78 cm. The labels-only margin then silently covers only 0.67 (bike) and 0.74 (plant). The joint margin stays valid (0.93, 0.96) but abstains in 60 and 86 % of the splits. In missions the uncalibrated planner finds a path in 23 % of the tasks, and 44 % of those paths are unsafe; the joint arm abstains in 89 % of the tasks. With this pose quality there is no useful certificate, and the method reports it instead of issuing a false one. Margins calibrated on ReplicaCAD and tested on HM3D scenes [21] of OSMa-Bench do not transfer: of the two scenes in which a plant is visible from the trajectory, the detector finds one (miss 0.18 m) and misses the other (2.9 m), so the guarantee has to be recalibrated where the robot operates.

VII. DISCUSSION AND LIMITATIONS

The joint margin is not better than label sets where their assumptions hold. It is better where they fail: under pose drift and with an open vocabulary. Two of our expectations were wrong. Positive correlation between the errors makes the sum of margins more conservative; under-coverage appears when rare failures of the two modules hit different scenes. And with 13 calibration scenes the worst scene sets the margin, so neither a bolder risk level nor better masks help much. Two attempts to shrink the margin failed: changing the region shape (convex hull, morphological closing) did not change the miss distance, and a margin scaled by the confidence of each detected fragment did not shrink the keep-out zone. The bottleneck is confident misses and class swaps of whole objects. If they occur in more than an α fraction of scenes, no class margin can certify that class, and map metrics such as mIoU do not show it; a “fraction of scenes with a whole-object miss” metric would fit mapping benchmarks.

Limitations: one family of 21 simulated scenes and one lighting condition; a box detector rather than a full mapper; synthetic drift, with real odometry tested only without loop closure; a guarantee that is marginal over scenes and holds in the planner frame at query time, not during execution; an out-of-distribution check on two scenes. Plant and bike were chosen as mission classes after seeing that the TV stand abstains; the three-class results are in the code repository.

VIII. CONCLUSION

The miss distance in the planner frame is one calibrated quantity that gives a planner a distribution-free guarantee against label and pose errors together. Margins calibrated on either error alone lose coverage under the other, and the sum of separate margins depends on an error dependence that is unknown in practice. The remaining price is set by the worst perception errors, which points to more calibration scenes, SLAM with loop closures and full open-vocabulary mappers as next steps.

REFERENCES

- [1] Q. Gu *et al.*, “ConceptGraphs: Open-vocabulary 3D scene graphs for perception and planning,” in *Proc. IEEE ICRA*, 2024, pp. 5021–5028, doi: 10.1109/ICRA57147.2024.10610243.
- [2] M. Popov *et al.*, “OSMa-Bench: Evaluating open semantic mapping under varying lighting conditions,” in *Proc. IEEE/RSJ IROS*, 2025, pp. 10786–10791, doi: 10.1109/IROS60139.2025.11247603.
- [3] D. Rotondi *et al.*, “3D scene graphs: Open challenges and future directions,” arXiv:2606.19383, 2026.
- [4] A. N. Angelopoulos and S. Bates, “A gentle introduction to conformal prediction and distribution-free uncertainty quantification,” arXiv:2107.07511, 2021.
- [5] D. S. Sundarsingh *et al.*, “Safe planning in unknown environments using conformalized semantic maps,” *IEEE Robot. Autom. Lett.*, vol. 11, no. 4, pp. 4921–4928, 2026, doi: 10.1109/LRA.2026.3668466.
- [6] Z. Mei *et al.*, “Perceive with confidence: Statistical safety assurances for navigation with learning-based perception,” *Int. J. Robot. Res.*, vol. 45, no. 6, pp. 938–967, 2025, doi: 10.1177/02783649251378151.
- [7] R. Baral, “Marginal calibration does not compose: Hidden dependence in modular robot navigation,” arXiv:2609.23731, 2026 (IROS 2026 workshop).
- [8] R. Kurkova, M. Popov, and S. Kolyubin, “OSMa-Bench++: Toward open-ended benchmarking of semantic mapping for manipulation with prompt-generated synthetic scenes,” arXiv:2605.26831, 2026 (ICRA 2026 workshop).
- [9] J. M. Correia Marques, A. Zhai, S. Wang, and K. Hauser, “On the overconfidence problem in semantic 3D mapping,” in *Proc. IEEE ICRA*, 2024, pp. 11095–11102, doi: 10.1109/ICRA57147.2024.10611306.
- [10] L. Lindemann *et al.*, “Safe planning in dynamic environments using conformal prediction,” *IEEE Robot. Autom. Lett.*, vol. 8, no. 8, pp. 5116–5123, 2023, doi: 10.1109/LRA.2023.3292071.
- [11] J. Shin, Y. Ra, and I. Yang, “From prediction uncertainty to conformalized distance fields for safe motion planning,” arXiv:2607.00776, 2026.
- [12] D. Kumar *et al.*, “Learnable conformal prediction with context-aware nonconformity functions for robotic planning and perception,” in *Proc. IEEE ICRA*, 2026, arXiv:2509.21955.
- [13] S. Natraj, B. Sinopoli, and Y. Kantaros, “Conformal constraint tightening for chance-constrained motion planning with unknown dynamics,” arXiv:2607.22409, 2026.
- [14] G. C. Calafiore, “Bridging conformal prediction and scenario optimization: Discarded constraints and modular risk allocation,” arXiv:2603.19396, 2026.
- [15] H. Sier *et al.*, “P-POSEMEM: Projective semantic memory for consistent language grounding under pose-graph rewrites,” arXiv:2609.15475, 2026.
- [16] A. Szot *et al.*, “Habitat 2.0: Training home assistants to rearrange their habitat,” in *Proc. NeurIPS*, 2021, arXiv:2106.14405.
- [17] T. Cheng *et al.*, “YOLO-World: Real-time open-vocabulary object detection,” in *Proc. IEEE/CVF CVPR*, 2024, pp. 16901–16911, doi: 10.1109/CVPR52733.2024.01599.
- [18] Q.-Y. Zhou, J. Park, and V. Koltun, “Open3D: A modern library for 3D data processing,” arXiv:1801.09847, 2018.
- [19] F. Steinbrücker, J. Sturm, and D. Cremers, “Real-time visual odometry from dense RGB-D images,” in *Proc. IEEE ICCV Workshops*, 2011, pp. 719–722, doi: 10.1109/ICCVW.2011.6130321.
- [20] C. Zhang *et al.*, “Faster segment anything: Towards lightweight SAM for mobile applications,” arXiv:2306.14289, 2023.
- [21] S. K. Ramakrishnan *et al.*, “Habitat-Matterport 3D dataset (HM3D): 1000 large-scale 3D environments for embodied AI,” in *NeurIPS Datasets and Benchmarks*, 2021, arXiv:2109.08238.